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PRACTITIONER'S PERSPECTIVE

Applied autoethnography: A method for reporting best practice in ecological and environmental research

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Abstract

1. Applied research involves interactions between different organisations—academia, industry, government. Breakdowns in communication can occur during these interactions which alter a project's outcome. We omit how we encounter and overcome these problems from scientific manuscripts which mask the social and cultural considerations that are critical to a project's success.
2. Autoethnography is a form of structured reflection whereby researchers use personal experience to contribute to understanding collaborative processes. We propose an applied form of autoethnography as a repeatable protocol to describe inter-organisational interactions during the research process in ecology and environmental research. We demonstrate the use of this protocol with five case studies from a diversity of wildlife research across a wide variety of experience levels and scales from small mammals, large herbivores and predators to digital ecology.
3. Our applied autoethnography protocol would ensure that specific biases and context are adequately described and that problems encountered and lessons learned from the experience are reflected upon.
4. These reports can be presented as stand-alone publications where appropriate, that is, to communicate an effective solution for a novel problem, or within the methods or supplementary material of manuscripts to further explain how the project developed from initial idea to final publication. Furthermore, this protocol can be used by practitioners to evaluate the trajectory of management decisions and policy implications in their jurisdiction to promote transparency and improve communication with stakeholders.
5. *Synthesis and Applications*: Applied science will continue to intersect with organisations that help or hinder research efforts depending on cultural contexts and biases. Using adequate reflection on case studies to record these experiences and disseminate lessons to the wider community will improve how we approach problems in research, help us to avoid repeating mistakes and ultimately save time and resources. Outside of research, case studies derived from

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this protocol allow practitioners to holistically understand the methods, biases and challenges of the research from a new perspective, thus providing a novel knowledge brokering function between academia and practitioners in applied ecology.

KEYWORDS
applied ecology, autoethnography, collaboration , data collection, methods in ecology, multidisciplinary research, stakeholder communication, transdisciplinary methods, wicked problems in ecology

1 | INTRODUCTION

Research is iterative, messy and ideally collaborative, yet its true nature is belied by the neat presentation of structured scientific manuscripts (Muncey, 2005). Medawar (1990) argued the structure of scientific manuscripts veils many actions which occur between observation and final publication. We argue that ecologists should communicate what happens behind this veil. In applied fields that involve multiple stakeholders—ecology among them—researchers dedicate time to building relationships, understanding perspectives and navigating controversy. Our continued failure to explicitly report on these experiences in our writing means future researchers will not fully comprehend the methods of how work was carried out.

In applied ecological and environmental research, these experiences are nuanced, with inherent cultural, societal and professional norms that impact the success of research. Every research project has to navigate, for example, the different organisational cultures between academia and private industry or the cultural differences when interacting with organisations from a different country. How best to collaborate with stakeholders and professionals with different backgrounds requires careful consideration. However, in applied ecological and environmental research, we often neglect to record how we established collaboration and how we acquire specialised background knowledge.

In Figure 1, we visualise a sample of factors which can influence a research project. Many of the relationships between factors are dynamic with changing agendas, viewpoints and biases dependent on the researcher and the nature of these interactions between factors. For instance, consider the recent surge of interest in bats as disease vectors in light of the COVID-19 pandemic which now renders this group far more controversial, or lucrative to study depending on the nature of the project with consequences for interactions with the public, the media and the study species.

We propose autoethnography as a method in ecological and environmental research to establish an evidence base of these experiences. Autoethnography is a form of qualitative research whereby the researcher uses personal experience to contribute to a wider field of cultural, social and scientific understanding (Ellis et al., 2011). This method was established in the field of anthropology but is widely used in many fields for instance, medicine, psychology, business, education and sociology but is, as yet, not used in ecology and environmental science.

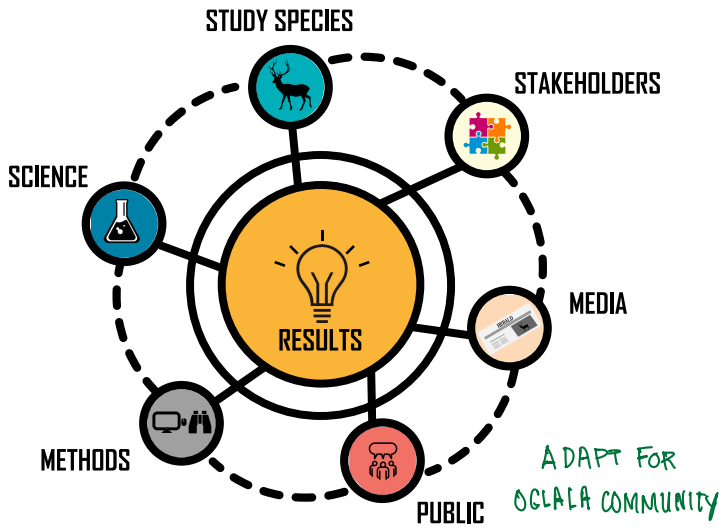


FIGURE 1 Different factors which can affect the outcome of research. Applied research often features at least two of these factors, and the characteristic of these factors can alter the results of the project e.g. elusive study species, negative media attention, poor methodological decisions; all of which can impact the outcome of the project. These factors can directly (solid lines) or indirectly (dashed lines) affect the process of research through interactions with each other and/or with the research. It is important to note, factors displayed here were chosen based on the case studies presented in the manuscript, other factors note featured here may be present in other fields of ecological and environmental research and should be highlighted in future study cases

As applied ecology works to tackle so-called 'wicked' problems, that is, those issues that encompass social, economic and political considerations, we believe autoethnography will become increasingly valuable. By reporting the obstacles, methods and solutions that researchers encountered within the human dimensions of their projects, we can avoid repeated mistakes and save valuable time. Furthermore, applied research is beset by the compartmentalisation of scientific fields, professional practices, sector-based division of responsibility and lack of widespread successful collaboration between scientists, policymakers and the public (Brown et al., 2010). The trend towards quantitative methods in ecology and environmental research may widen this gap as resulting scientific insights are

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not inherently digestible unless disseminated effectively (Fawcett & Higginson, 2012). Reporting on the human element of quantitative projects will add further context to the methods, increase transparency and improve the research application. This is not unique to quantitative methods, in fact, qualitative methods will also benefit from the reflective process and improve how researchers communicate transparently to a wider audience.

The rapid development of transdisciplinary approaches to tackle problems around sustainability indicates that many practitioners are realising the importance of knowledge co-production from the outset (Roux et al., 2017). There is room for improvement. For instance, research in the field of land-use science found that key elements of transdisciplinary research were lacking from applied projects, for example, collaboration between science and industry, interdisciplinary teams and the lack of coordination for policy and management (Zscheischler et al., 2017). Norström et al. (2020) demonstrate how complex multidisciplinary problems are best tackled collaboratively between academics and non-academic organisations in sustainability science despite the challenges of knowledge co-production between researchers and practitioners. Among ecologists, we argue that as we begin to reflect on the human problems encountered in our research, there will be a further push to explore transdisciplinary literature and methods. Failure to understand the theoretical foundations of these fields may lead to reduced validity in study design, thus ensuring the emergence of problems during the research process or a misinterpretation of the findings as a result of failing to holistically evaluate the research (Moon & Blackman, 2014). It is therefore imperative that ecologists begin to incorporate other fields that influence their research (see Moon & Blackman, 2014 for a detailed guide on understanding the social science which may underpin applied ecological research). Thus, we argue that ecologists must begin to incorporate these methods into our own discipline and report on the outcomes of these facets of our research.

In this paper, we describe our adapted version of autoethnography, which we call applied autoethnography, as a standardised protocol for researchers to report on methods and practices that stem from the human elements of research. Our protocol is an accessible way for researchers to document their collaborative approaches, problems and lessons for other practitioners. Lack of experience navigating transdisciplinary problems in ecology means the community is destined to repeat mistakes, resulting in wasted time and resources. Previous studies have called for increased research and reporting on inter/intranetwork communication and problem-solving in transdisciplinary applied research (Pade-Khene et al., 2013). Our protocol fills this research gap and aims to generate an evidence base of previously veiled scientific methods.

2 | APPLIED AUTOETHNOGRAPHY

Autoethnography aims to put into context the experience of the researcher within the natural and social world they inhabit and the subcultures they define themselves by Miles (2017). The practice of autoethnography was introduced by transdisciplinary researchers, for example, Carolyn Ellis and Arthur Bochner, who encouraged experimentation in its use (Anderson, 2006). Despite this, there are challenges associated with using this method correctly (see Forber-Pratt, 2015) in academic research that must be carefully considered. To adapt this method for ecological and environmental research, we propose that researchers reflect on and report the following five aspects of their research:

1. Non-technical project details which were instrumental in project establishment and success. We define non-technical project details as methods employed by the researcher to navigate the human element of applied research, for example, accessing private field sites, collecting data from stakeholders, etc. Furthermore, the gathering of informal information is essential to understand the trajectory of the research decisions taken. Certainly, good ethical practice should be understood with respect to the recording, dissemination and distribution of practitioner knowledge and information within research. See (Minteer & Collins, 2005) for a practical understanding of ethical considerations in applied research scenarios.
2. Regional & cultural bias. Reporting on and understanding the researcher profile (gender, age, nationality, ethnicity, language fluency) allow future researchers to understand the context of decision-making and evaluate the usefulness for their own research. This may not be a straightforward procedure and careful consideration and training into implicit bias should be undertaken prior to reporting these facets of the research (see Kim and Roberson (2021) for a review on training to recognise implicit bias and report it, also see Bowser and Cid (2021) for an application in ecology). These biases can also be discussed in the historical and political context of researchers working in foreign countries or within multinational teams. This encompasses biases regarding communication, tool use, protocol design, etc. (i.e. not only problems that arose due to the individual age–sex identity being met with questions, but also biases with respect to methodology selection and study design).
3. Spatial & temporal scales. Acknowledging where research took place, at what scale and the time involved to provide important contextual information. Note this should encompass holistic reporting of the spatial and temporal scales, rather than a duplicate of a methods section it should report on the full extent of the research from project inception to completion, for instance discussions with stakeholders, scoping processes, data accessibility discussions whether successful or unsuccessful that were conducted prior to the collection of the project data.
4. Problems encountered within the project, providing key lessons learnt through failure and good practice learned through success. Documenting problems encountered will create a shared foundation for future research on similar problems and will aid the iterative improvement of methods.
5. Lessons for practitioners, whereby the lessons learned are recorded and can be assessed for similar projects in the future.

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We used these five components of autoethnography and incorporated them into a standardised protocol for writing a structured repeatable case study (see Supplementary Material S1 for full writing protocol and see Figure 2 for a graphical representation of the protocol). By using this protocol, we demonstrate how to communicate the reasoning behind decision-making and collaborative approaches. Our protocol emphasises problem-solving and community learning by promoting reflection on specific problems encountered by researchers, the solutions devised and the lessons learned from the experience. Our applied autoethnography is more concise, standardised and repeatable than traditional autoethnography by focusing on specific lessons learned with the aim to create an evidence base of problems and solutions within a structured protocol (Ellis et al., 2011).

The reports written via our protocol can be presented as standalone publications after the initial publication where appropriate, for example, a new method is found to traverse a novel problem, or within the methods or supplementary material of quantitative projects to contextualise the data collected/methods employed or where interactions between factors (Figure 1) may have influenced the results of the study. If located in the methods section of a paper, care should be taken not to disclose information which has the potential to remove objectivity and introduce bias for reviewers. If located within the supplementary material, we advise researchers to include a sentence in the methods section highlighting that they included an autoethnographic account to communicate issues encountered, similar to the inclusion of an ethics statement. These reports can also be published as grey literature for multidisciplinary teams to learn from the different perspectives of team members working on the same problem. The proposed protocol delivers a standardised and repeatable approach to autoethnography to fit within the researchers' toolbox to act as a knowledge brokering tool for community learning.

We present five examples from wildlife biology using our repeatable protocol, see Supplementary Material S1 and Table 1. These case studies are representative across academic level (early career researcher through to senior academic) highlighting that the issues encountered here are not associated with a particular level of experience, rather, that these problems arise from a lack of understanding and awareness for multidisciplinary methods which are increasingly becoming integrated in applied ecological research (Moon & Blackman, 2014). Below, we present our in-text case study which details an early career researcher working in a complex polarised socio-ecological landscape. This individual worked as part of an academic team, supervised by established researchers who also failed to foresee the problems encountered. Thus, this case study encompasses a spectrum of age–sex demographics, experience levels and cultural contexts in a single case study.

3 | IN-TEXT CASE STUDY. UNDERSTANDING LOCAL PERCEPTIONS OF PREDATORS IN A FOREIGN AND POLARISED ENVIRONMENT

3.1 | Project title

Investigating drivers of tolerance towards dingoes and wild dogs in Queensland, Australia.

3.2 | Researcher profile

Mid-twenties, female, British researcher. During this study, I was completing my MSc in Conservation Science. Previous to this project, I had been involved in small-scale research projects interviewing members of the public; however, this was the first research project which I had led myself, and my first experience carrying out research in a country outside of the United Kingdom. Note, while this project was conducted as an early career researcher, supervision was provided by experienced and senior academics including an Australian male and a South African female researcher. Both had experience conducting research in similar contexts using the wildlife tolerance model but did not to foresee the challenges encountered in this project.

3.3 | Context

Human–wildlife conflict is a global problem. However, the way that these conflicts present themselves, and the manner in which people

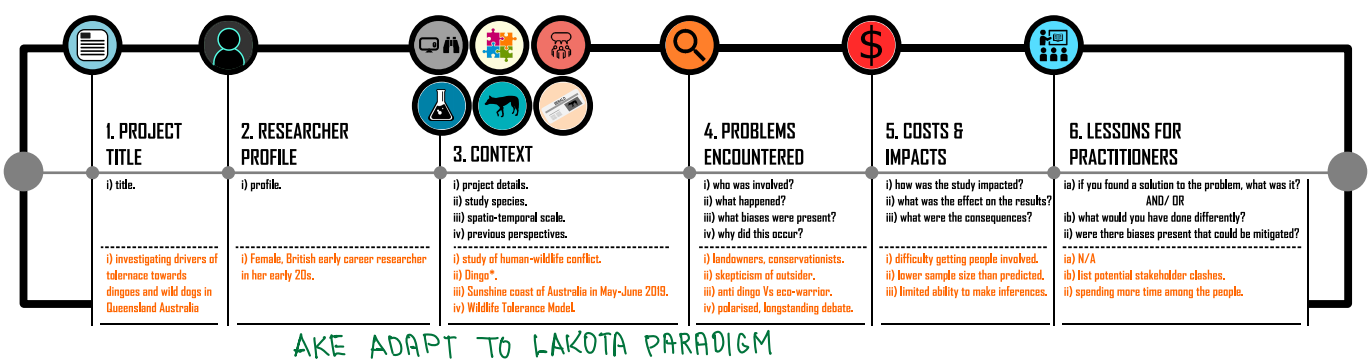


FIGURE 2 Graphical application of our structured applied autoethonography case study protocol. Examples of this in applied research are featured in-text and in supplementary material. Key features are noted in black text and applied examples from our in-text case study are demonstrated in orange text. For the full protocol on how to write a repeatable case study report, please see supplementary material 1.
* Scientific name is not provided as there is no consensus on Dingo species name, a key problem described in our in-text case study

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TABLE 1 Example case studies from projects in wildlife biology with spatial scale ranging from field site to global digital workspace

Spatial scale	Researcher	Case study title	Reference	Published manuscript
Field site	Amy Haigh	Utilising local ecological knowledge for locating an elusive species	Supplementary material S2	Haigh, A., O'Riordan, R.M. and Butler, F. (2012). Nesting behaviour and seasonal body mass changes in a rural Irish population of the Western hedgehog (Erinaceus europaeus). *Acta Theriologica*, *57*(4), 321–331.
City Park	Laura Griffin	Implementing cultural considerations in signage design for human–wildlife conflict	Supplementary material S3	Griffin, L.L., Haigh, A., Conteddu, K., Andaloc, M., McDonnell, P. and Ciuti, S. (2022). Reducing risky interactions: Identifying barriers to the successful management of human–wildlife conflict in an urban parkland. *People and Nature*.
Regional	Grace Nolan	Understanding perceptions of predators in a foreign and polarised environment	In-text	NA
National	Simone Ciuti	Establishing a national stakeholder network and collating empirical data for a controversial species, the SMARTDEER project	Supplementary material S4	Morera-Pujol, V., Murphy, KJ., Burkitt, T., Coad, B., McMahon, B.J., Nieuwenhuis, M., Morelle, K., Ward, A., Ciuti, S., 2022. Bayesian species distribution models integrate presence-only and presence-absence data to predict deer distribution and relative abundance. bioRxiv.
Global digital workspace	Adam Kane	Natural history films raise species awareness—A big data approach	Supplementary material S5	Fernández-Bellon, D. & Kane, A., (2020) Natural history films raise species awareness—A big data approach. *Conservation Letters*, *13*(1), e12678.

tolerate and cope with problem-causing species differ on global, national and local scales. For my masters in conservation science, I decided to apply the Wildlife Tolerance Model (Kansky et al., 2016) to dingoes & wild dogs to understand what drives tolerance towards a controversial species that is considered both native and invasive with little consensus and implicated in high levels of human–wildlife conflict. I conducted 2 months of fieldwork (May & June 2019) covering the entire Sunshine Coast region of Queensland, Australia. The aim of this was to interview and survey local landholders, stakeholders and members of the public regarding their experience living alongside invasive species, with a focus particularly on dingoes and wild dogs. The majority of previous studies concerning dingoes focused on their ecological roles as apex predators and trophic regulators or problematic invasive species. However, recent work has begun to focus more on public perceptions of the dingo and management practices.

3.4 | Problems encountered

There is currently no scientific consensus on whether dingoes are their own species (*Canis dingo*), a subspecies of the domestic dog (*Canis familiaris dingo*) or just wild-living domestic dogs (*Canis familiaris*). This lack of consensus is also present in official government documents and legislation across Australia. In addition to the debate around species or subspecies, debate also rages over their status as native, invasive or naturalised, and whether they should be protected or persecuted. Whilst this was apparent from the literature review before embarking on my fieldwork, I was unprepared for how this debate (which I incorrectly assumed would be limited to academic discourse) would manifest in my communications with stakeholders and the general public.

Dingoes are a politicised issue. Whilst I was conducting my fieldwork, a programme of 1080 poison baiting was also being run by the local council pest management team and supported by Queensland Department of Agriculture and Fisheries. This is a highly controversial practice that targets the reduction of numbers of 'wild dogs' and foxes (invasive to Australia) in an area by leaving poisoned bait in areas of high wild dog activity. This ongoing 1080 baiting programme affected my ability to recruit stakeholders into my study. Some potential study recruits refused to participate upon finding out Biosecurity Queensland had given me permission to conduct this research believing that I was 'anti-dingo' and collecting evidence to further support 1080 baiting practices and kill 'native' dingoes. On the other side of the spectrum, I was accused of being an 'eco-warrior' out to protect 'invasive' wild dogs and allow the continued depredation of their livestock and livelihoods. Although my study was completely independent from the 1080 baiting, the high levels of controversy surrounding the practice meant potential participants were unable to separate their stance on 1080 baiting from my study and refused to participate.

Whilst the study was supported behind the scenes by the local council, they were unable to provide public support to the study. The

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lack of a strong affiliation with a local trusted organisation was damaging to the study. As a British researcher with no previous ties to the region I was met with suspicion by many potential participants, despite my supervisor being a senior Australian academic. People questioned why a university in the United Kingdom was interested in their experience living alongside dingoes in Australia. This was particularly evident when approaching more sensitive questions such as economically quantifying loss experienced due to depredation, or number of livestock kept on the property. Many participants did not wish to disclose any financial information despite reassurances (verbal and written) that data were anonymous and would not be made available to any third parties. This meant that aspects of the study that dealt with quantifying loss and understanding stakeholder opinions towards potential insurance and compensation programmes were skewed or incomplete.

Similarly, as an outsider who has not shared the same lived experience including devastation to their livestock or strong emotional responses to the culling of 'native' dingoes many people I encountered questioned my credentials and whether I was equipped to understand the complexity of the situation. In addition to this, a lack of understanding of the stakeholder demographics meant the original plan to recruit a large portion of the study participants online and collect surveys online was flawed. Many farmers in the more rural areas were not very computer literate or did not have strong internet connections.

It became increasingly evident throughout this study that there were high levels of discontent in the community with the current management of dingoes and wild dogs and huge polarisation within the community. Some believed not enough was being done, and others believed the current plans were barbaric or damaging to native wildlife. The issues I had gone to study, tolerance towards dingoes and wild dogs, were far more complex and deep rooted in politics than I could have anticipated.

3.5 | Costs and consequences

Two months was not a long enough period to conduct this fieldwork. The wildlife tolerance model has been successfully deployed in other contexts around the world and naively we assumed that this context would be no different. However, the extent of the political complexity surrounding dingoes and wild dogs had been severely underestimated. As a result of this, I was unable to collect the number of surveys and interviews required to complete the analysis using the methods originally planned.

3.6 | LESSONS TO PRACTITIONERS

Speaking the same language as someone does not mean that you will easily be able to conduct research. We assumed that as there would be no language barrier, it would be easy to form relationships with participants and recruit people into the study. We also assumed that people would be willing to discuss their experiences. This may have been the case, but as I was not a local and had not lived a shared experience people were not always willing to speak to me, a British female, with no agricultural background.

Strong relationships with stakeholder representative groups such as environmental NGOs and agricultural unions should have been established too. Reliance on the relationships established with the Department of Agriculture and Fisheries or the Sunshine Coast Regional Council was naïve as these departments were unable to offer public support to the study and there was a feeling of distrust towards them from many participants. A larger network may also have combated the issues encountered due to the controversial 1080 baiting practices and the assumption this study was associated with this.

Finally, and most importantly, we underestimated the complexity of the dingo and wild dog controversy. Whilst it was clear in the scientific literature, it could not have been anticipated how passionate the local landholders and general public felt about the status of the dingo (native or invasive? Distinct species or subspecies? Protect or persecute?). The passionate response from the local landholders on this subject suggested that preliminary studies to understand local perceptions of and ideas about dingoes and wild dogs should have been carried out before we attempted to conduct a Wildlife Tolerance Model study. In contexts where politics play such a major role in understanding public attitudes towards a species, a much more thorough scoping period is required before designing a study and starting fieldwork.

4 | DISCUSSION

Research projects tend to evolve through iterative communication, methods and practices until such time as we deem the results to be worth communicating with our peers. Once the research methods come to an end, they are replaced by story-telling, typically in the form of scientific manuscripts, presentations and media communication (Miles, 2017). Our case studies underscore the valuable lessons lost in the interim. Autoethnography encourages us to reflect on every aspect of our research. By doing so we can reveal and refine methods that are developed through ad hoc decision-making during research (Muncey, 2005). This provides our peers with a more complete methodology and contextual background to the study which will greatly improve how we report our science and interpret our results, for example, by understanding issues such as data availability, study site access or reasons that led to issues in working with stakeholders.

This is evident when comparing our case studies with the methods included in their published form (Table 1). Patterns emerge even within our small yet diverse sample. Many of the difficulties we encountered would have been resolved or at least minimised if we had taken a transdisciplinary approach in our research from the beginning. This speaks to the value of autoethnography because these hidden problems are now clear to us as well as our peers. Future

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efforts now have a head start and can learn from the mistakes made explicit here. Establishing an evidence base in the literature across ecological and environmental science will provide a record for future practitioners to refer to for applied research. Examining this evidence base with the nuance of contextualisation (e.g. understanding the researcher profile, regional context, scale) can help researchers adopt or adapt methodologies based on their understanding of their own context. This will aid future researchers to avoid repeating mistakes, saving time, resources and money.

Autoethnography is an optimal methodology to establish this evidence base as it can describe the objective lessons from subjective experiences, for example, through reflection and contextualisation the researcher can transform experience into applicable and repeatable methods for novel problems. When reporting quantitative research, it is best practice to remove "the self" from the methods in order to reduce bias and ensure repeatability. Perhaps, the opposite is true when reporting on facets of research which are not chosen but rather evolve through communication and human interaction. We stress that as ecology becomes ever more quantitative, it should not lose sight of these qualitative features—methods begin with people rather than data thus reflecting on these aspects of research will aid a holistic understanding of the research as we work on more projects outside of our traditional disciplines. In truth, many of the human interactions that occur within qualitative and quantitative research are data, as they inform our approach, methods and interpretation of the research.

Applied research does not exist in a vacuum which can result in subjectivity permeating the research methodology (e.g. denied access to a field site, ignored by important stakeholders, refusal to share data). This is a theme evident in all of our case studies, we argue that rather than ignore the subjectivity that emerges within research, we must develop better methods to understand it. The individual identity is unquestionably worthy of research as the narrative should represent all features of the experience which legitimise the reasoning behind the lessons for practitioners (Muncey, 2005). These issues can alter the outcome of the research as important data, stakeholders and other information sources may not be represented in the final analysis. Acknowledging this and reporting on success and failure through autoethnographic reports can contextualise the final research output and provide a framework for future research efforts in similar fields. This framework will facilitate future researchers to enter multidisciplinary projects with an evidence-based approach and will greatly improve collaborative efforts between researchers and policymakers in the future.

It is important to acknowledge the ethical considerations of the method we propose here. Firstly, some of the information provided by the authors might be sensitive and have repercussions for the reputations of professionals and stakeholders, or incur risk for institutional and personal relationships. We believe that authors reporting their autoethnography analysis must document their experiences honestly but they also need to be cautious to avoid future conflicts with collaborators and stakeholders. We offer a way for researchers to report the critical issues encountered during their research, so that future researchers will not repeat the same mistakes, but we do not suggest that it be made mandatory to disclose such information—authors, in some cases, may decide to withdraw their consent to disclose any details about the difficulties encountered along the way to avoid compromising their future work.

It is important to ensure that people's identities are preserved—remembering that individuals and organisations in small communities do not need to be named to be identified. In this case, European General Data Protection Regulation (GDPR) laws can act as a guidance document to avoid unethically and, unlawfully in some instances, disclosing people's personal information. Care must be taken to disclose methodological errors that were made during the research process. Unfortunately, it is not universally accepted that learning through experience is a positive goal—in the case of early career researchers, there are power dynamics at play that may risk damage to their reputations should they disclose errors. Similar to the other ethical issues raised above, researchers should run their autoethnography analysis and, before full disclosure, consider a risk analysis with collaborators in order to identify the risks connected to it. If the risks cannot be afforded, then the authors should add a statement to their papers where they declare they have considered an autoethnography assessment but decided not to disclose it due to ethical concerns.

Our protocol for applied autoethnography, adapted for ecological and environmental research, captures the full context of the research when supplementary to quantitative scientific manuscripts or as stand-alone reports for pilot studies. Additionally, our protocol also benefits the community by highlighting other valuable components of research and challenges such as a lack of interorganisational understanding between researchers, stakeholders and government in our in-text case study, uncovering biases that affect research success (Supplementary Material S3) and presenting new research directions, for example, scoping studies to investigate stakeholder perceptions prior to large-scale ecological studies (Supplementary Material S4). Whilst this protocol produces an output which can be used to understand the context of the research, it also has information embedded within each unique case study that will contain information that is tangible and usable in other contexts, thus bolstering the advantages of this method in applied ecology. Our case study protocol describes the researcher, the place and the circumstances of the study which can mould experiences into methods and best practices for researchers seeking to begin a new project or expand on existing efforts. As stand-alone reports, they offer detailed narratives on the choices taken as a result of human interaction and the outcomes for the research, for better or worse. As supplementary material to quantitative papers, these reports can wholly explain the context of data collection and data availability, describing the biases which shaped the analysis and the effect on the results. Communicating this via our standardised protocol and subjected to peer review will create an evidence base of human problems which

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ecologists encounter, allowing us to develop methods and practices to overcome them as a community and improve how we work with other organisations.

5 | CONCLUSION

The field of ecology and environmental science will continue to intersect with the public, policymakers and other organisations. This will help or hinder research efforts depending on their beliefs, cultural context and biases towards the project. Using autoethnographic case studies to record these experiences, learn from them and then disseminate these lessons to the wider community will improve how we approach problems with organisations outside of the academic community in applied research scenarios. As research becomes more applied, it is essential that researchers endeavour to provide evidence that improves practices for working with practitioners who seek to tackle these issues (Milner-Gulland et al., 2012). Additionally, these case studies highlight that science is socially constructed, such that applied research aims to inform policy, science planning, government decision-making, institutional reform and applied ecological practice. Our applied autoethnographic protocol can contribute to improve the application of our research through the holistic presentation of researchers' experiences.

It is imperative that we study applied fields (e.g. sustainability science, natural resource management) whose research involves a subset of the factors in Figure 1 (i.e. the public, stakeholders, the media) to learn from established practitioners how they navigate wicked problems. Transdisciplinary research (a research strategy that crosses many disciplinary boundaries to create a holistic approach) literature should also be explored prior to engaging in multidisciplinary research and methods in this field should be incorporated into ecological and environmental projects where appropriate. Learning from these fields and reflecting on our own experiences during our research will help create a holistic understanding of working in multidisciplinary research, improve our methods and improve how we collaborate in research. Human interactions occur in practically every research project in applied ecology and environmental science. Applied autoethnography will help our community to record and learn from these experiences, thus transforming past mistakes into future opportunities and squandered resources into new assets for more transparent and higher impact applied research, in addition to the many other benefits which we highlight throughout this paper outside of evaluation and learning through our structured protocol.

AUTHORS' CONTRIBUTIONS

K.J.M. contributed in conceptualisation and writing—original draft; K.J.M., L.L.G., G.N. and A.K. contributed in methodology; K.J.M., L.L.G., G.N., A.H., T.H., S.C. and A.K. contributed in writing—review and editing; A.K. contributed in visualisation and supervision.

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CONFLICT OF INTEREST

The authors have no conflict of interest to declare for this project.

DATA AVAILABILITY STATEMENT

All data for this project are qualitative and are included in the main text and supplementary material as case studies. Please see Table 1 for full list of the case studies.

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BIOSKETCH

Kilian J. Murphy Kilian's research focuses on providing data-driven solutions to wildlife management conflicts, particularly in data sparse environments. Kilian is currently working on establishing Ireland's first nationally coordinated data-driven monitoring scheme for large herbivore Ireland, working with a suite of relevant stakeholder and policymakers in the process. Research completed by this author includes management solutions for predator conflict, agricultural disease and energy development disturbance. These works, while data-driven, all included significant interaction with stakeholders, policymakers, the media and the public to understand the conflict and the context of current thinking on these issues among stakeholders. This intersection of researchers and practitioners facilitated education and contextual understanding, which bolstered all results obtained via statistical analysis. These interactions which have proved pivotal to project success are under-reported in ecology and environmental science which led to the inception of this project to change that.

Laura Griffin Laura's work focuses on the field of human–wildlife interactions, conflict and management, with particular emphasis on exploring ecological, behavioural and welfare implications for targeted wildlife populations. Laura is currently working on a project exploring the impacts of unregulated, visitor feeding interactions on wild fallow deer in a public park. This project involves close collaboration between herself, associated researchers and stakeholders as they focus on public communication and education. This author has previously been involved with research into the impacts of tourism activities on wildlife conservation with a focus on African elephants and protected sea turtle nesting sites. Each of these projects has involved communication across different invested parties and laypeople, without which project success would not be possible.

Grace Nolan Grace's research is focused on human–wildlife conflict and the impacts of conflict and conservation narratives on public perceptions of wildlife. Grace's current work is investigating media portrayals of human–wildlife conflict through print, digital, visual and social media. This work looks to identify how the media can influence public and stakeholder understandings of 'problem-causing' species and identify methods for scientists and conservationists to better collaborate with the media to mitigate any negative perceptions caused by human–wildlife interactions. Grace's previous research investigated drivers of tolerance towards dingoes and wild dogs in Queensland, Australia, which involved working directly with a variety of stakeholders, including local and state government, agricultural landowners and private protected area stewards.

Amy Haigh Amy has worked in University College Dublin since 2018, where her work has been mainly focused on assisting with research on fallow deer. Prior to this, she completed a post doc in Teagasc in the welfare department, investigating tail and ear biting in intensively farmed pigs. This project had a major stakeholder component and involved farm trials and surveys. From 2007 to 2014, her research (PhD and postdoc) was concentrated in the area of mammal ecology and behaviour, with hedgehogs and squirrels as the study species. Both of these projects involved working with landowners and the public. Prior to this, she completed research on human interactions with free-ranging ring-tailed lemurs.

Adam Kane Adam has been a lecturer in ecology at University College Dublin since 2018. He has a background in zoology and science communication. He has studied a range of species with a particular focus on scavengers and seabirds which involved fieldwork and engagement with a diversity of stakeholders. His current research focuses on animal movement and human–wildlife conflict. His recent work on media portrayals of wildlife recognises that conservation needs to focus on the human element to effect behavioural change. Through his study of science communication, he has come to appreciate that 'the public' cannot be easily grouped together.

Tamara Hochstrasser Tamara is a lecturer in Sustainability Science at UCD, where she teaches on knowledge integration at the science–practice interface. An ecologist by training, her research is answering practitioner lead questions in transdisciplinary teams. In her work, she has provided evidence for decision-making for the management of urban greenspaces, advice on agricultural mitigation and adaptation to climate change, soil moisture monitoring and vegetation management in urban, rural and coastal settings. She is a policy representative for the Irish Ecological Association and the initiator of the ECOBROKER research question exchange platform.

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Simone Ciuti Simone is a quantitative behavioural ecologist interested in studying wildlife ecology and behaviour within human-dominated landscapes and tackling human–wildlife conflicts. Simone is currently leading the Smartdeer project, Ireland's first nationally coordinated initiative to provide evidence-based management solutions for large herbivores to stakeholders and policymakers. This pilot study involved building a diverse stakeholder network working towards better monitoring for polarised species. Simone has worked with stakeholders on human-wildlife conflict projects for a variety of species across the world, garnering an abundance of experience in balancing communication, collaboration and implementation of scientific data and policy.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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